6.Network Design and Management

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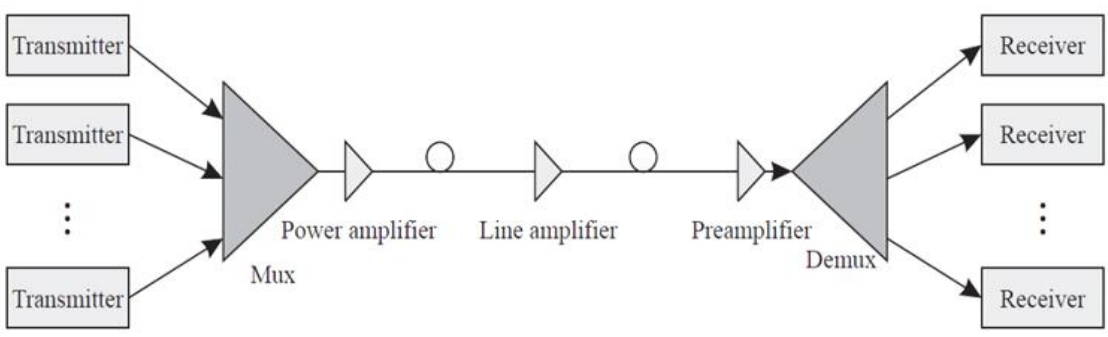
Transmission System Engineering

- Design the physical layer
- Allocate power margin for each impairment
- Make trade-off

System Model:

- Figure shows a block diagram of the various components of a unidirectional WDM link.
- The transmitter consists of a set of DFB lasers, with or without external modulators, one for each wavelength.
- The signals at the different wavelengths are combined into a single fiber by means of an optical multiplexer.
- An optical power amplifier may be used to increase the transmission power. After some distance along the fiber, the signal is amplified by an optical in-line amplifier.
- Depending on the distance, bit rate, and type of fiber used, the signal may also be passed through a dispersion-compensating module, usually at each amplifier stage.
- At the receiving end, the signal may be amplified by an optical preamplifier before it is passed through a demultiplexer. Each wavelength is then received by a separate photodetector.



- The physical layer of the system must ensure that bits are transmitted from the source to their destination reliably.
- The measures of quality are the bit error rate (BER) and the additional power budget margin provided in the system.
- Usually the required bit error rates are of the order of 10—9 to 10—15, typically 10—12. The BER depends on the amount of noise as well as other impairments that are present in the system.
- The non-return-to-zero (NRZ) modulation is used. In some specific cases, such as chromatic dispersion, both NRZ and return-to-zero (RZ) modulation are considered

Power Penalty

- The physical layer design must take into account the effect of a number of system Impairments.
- Usually, each impairment results in a power penalty to the system. In the presence of an impairment, a higher signal power will be required at the receiver in order to maintain a desired bit error rate.
- One way to define the power penalty is as the increase in signal power required (in dB) to maintain the same bit error rate in the presence of impairments.
- Another way to define the power penalty is as the reduction in signal-to-noise ratio as quantified by the value of γ due to a specific impairment.

Let P_1 denote the optical power received during a 1 bit, and P_0 the power received during a 0 bit without any system impairments. The corresponding electrical currents are given by P_1 and P_0, respectively, where is the responsivity of the photodetector.
Let σ_1 and σ_0 denote the noise standard deviations during a 1 bit and a 0 bit, respectively. Assume that the noise is Gaussian. The bit error rate, assuming equally likely 1s and 0s.

$$\mathrm{BER} = Q\left(\frac{\mathcal{R}(P_1 - P_0)}{\sigma_1 + \sigma_0}\right).$$

In the presence of impairments, let P_1^r, P_0^r, σ_1^r, σ_0^r denote the received powers and noise standard deviations, respectively. Assuming an optimized threshold setting, the power penalty is given by

$$\mathrm{PP} = -10\log\left(\frac{\frac{\mathcal{R}(P_1'-P_0')}{\sigma_1'+\sigma_0'}}{\frac{\mathcal{R}(P_1-P_0)}{\sigma_1+\sigma_0}}\right).$$

Case 1: Thermal Noise
The first case of interest is when the dominant noise component is receiver thermal noise, for which $\sigma 0 = \sigma 1 = \sigma$th. This is usually the case in unamplified direct detection *pin* receivers.
In this case, or in any situation where the noise is independent of the signal power, the power penalty is given by

$$\mathrm{PP}_{\text{sig-indep}} = -10\log\left(\frac{P_1' - P_0'}{P_1 - P_0}\right)$$

Case 2: Shot Noise

The other case of interest is amplified systems, or systems with APD receivers. In amplified systems, the dominant noise component is usually the amplifier signal- spontaneous beat noise.
In APD receivers, the dominant noise component is the shot noise, which is enhanced because of the APD gain In amplified systems, and in systems with APD receivers
In amplified systems, and in systems with APD receivers, we can assume that σ1 ∝ √P1; that is, the noise variance depends on the signal power. Assume also that P0 P1. In this case, we can assume that σ1 σ0. Here an optimized receiver would set its threshold close to the 0 level.

$$\mathrm{PP}_{\text{sig-dep}} = -5 \log \left(\frac{P_1'}{P_1} \right).$$

- System design requires careful budgeting of the power penalties for the different impairments.
- First we determine the ideal value of the parameter γ that is needed. For a bit error rate of 10^{-12} typically assumed in high-speed transmission systems, we need $\gamma = 7$, or $20 \log \gamma = 17$ dB.
- This would be the case if there were no transmission impairments leading to power penalties. In practice, the various impairments result in power penalties that must be added onto this ideal value of γ , as shown in Table , to obtain the required value of γ that the system must be designed to yield.
- For instance, in the table, we allocate a 1 dB power penalty for an imperfect transmitter and a 2 dB power penalty for chromatic dispersion

Impairment	**Allocation (dB)**
Ideal γ	17
Transmitter	1
Crosstalk	1
Chromatic dispersion	2
Nonlinearities	1
Polarization-dependent loss	3
Component aging	3
Margin	3
Required γ	31

Transmitter

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The key system design parameters related to the transmitter are its output power, rise-/fall-time, extinction ratio, modulation type, side-mode suppression ratio, relative intensity noise (RIN), and wavelength stability and accuracy.

(i) Output Power

- The output power depends on the type of transmitter. DFB lasers put out about 1 mW (0 dBm) to 10 mW (10 dBm) of power.
- An optical power amplifier can be used to boost the power, typically to as much as 50 mW (17 dBm). The upper limits on power are dictated by nonlinearities and safety considerations

(ii) Extinction Ratio

- The extinction ratio is defined as the ratio of the power transmitted when sending a 1 bit, P_1, to the power transmitted when sending a 0 bit, P_0.

$$r = \frac{P_1}{P_0}$$

$$P_1: \text{ power of bit "1"}, \quad P_0: \text{ power of bit "0"}$$

$$\text{For ideal case, } P_0 = 0, \quad r = \infty$$

$$\text{average power } P = \frac{1}{2} P_1$$

- This would correspond to an extinction ratio $r = \infty$. Practical transmitters, however, have extinction ratios between 10 and 20. With an extinction ratio r, we have

$$P_0 = \frac{2P}{r+1}$$

□ and

$$P_1 = \frac{2rP}{r+1}.$$

- Reducing the extinction ratio reduces the difference between the 1 and 0 levels at the receiver and thus produces a penalty. The power penalty due to a nonideal extinction ratio in systems limited by signal-independent noise is

$$\mathrm{PP}_{\text{sig-indep}} = -10 \log \frac{r-1}{r+1}.$$

- On the other hand, if we assume that the two systems have the same peak transmit power, that is, the same power for a 1 bit, then the penalty can be calculated to be

$$\mathrm{PP}_{\text{sig-indep}} = -10 \log \frac{r-1}{r}.$$

Receiver

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- The key system parameters associated with a receiver are its sensitivity and overload parameter.
- The sensitivity is the average optical power required to achieve a certain bit error rate at a particular bit rate. It is usually measured at a bit error rate of 10 −12 using a pseudorandom 223 − 1 bit sequence.
- Typical sensitivities of different types of receivers for a set of bit rates are shown in Table APD receivers have higher sensitivities than pin FET receivers and are typically used in high-bit-rate systems operating at and above 2.5 Gb/s. However, a pin FET receiver with an optical preamplifier has a sensitivity that is comparable to an APD receiver.
- The overload parameter is the maximum input power that the receiver can accept.
- The overload parameter defines the dynamic range of the receiver and can be as high as 0 dBm for 2.5 Gb/s receivers, regardless of the specific receiver type.

Bit Rate	Type	Sensitivity	Overload Parameter
155 Mb/s	pinFET	−36 dBm	−7 dBm
622 Mb/s	pinFET	−32 dBm	−7 dBm
2.5 Gb/s	pinFET	−23 dBm	−3 dBm
2.5 Gb/s	APD	−34 dBm	−8 dBm
10 Gb/s	pinFET	−18 dBm	−1 dBm
10 Gb/s	APD	−24 dBm	−6 dBm
40 Gb/s	pinFET	−7 dBm	3 dBm

Optical Amplifiers

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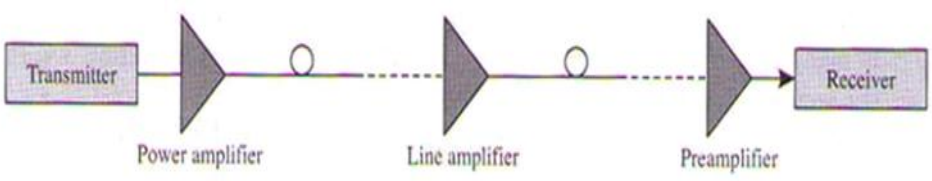
- Optical amplifiers have become an essential component in transmission systems and networks to compensate for system losses.
- The most common optical amplifier today is the erbium-doped fiber amplifier (EDFA) operating in the C-band. L-band EDFAs and Raman amplifiers are also used.
- EDFAs are used in almost all amplified WDM systems, whereas Raman amplifiers are used in addition to EDFAs in many ultralong-haul systems.



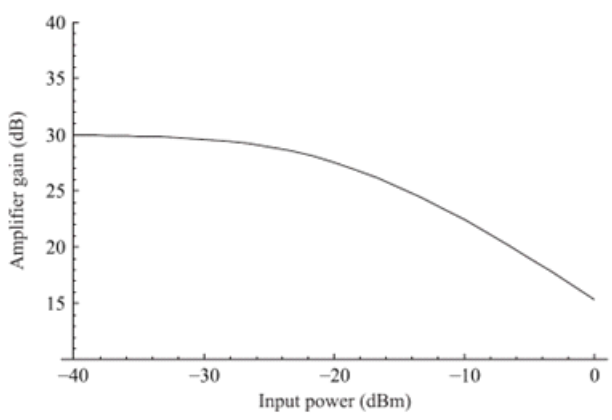
A power amplifier is designed to provide the maximum possible output power. A preamplifier is designed to provide high gain and the highest possible sensitivity, that is, the least amount of additional noise. A line amplifier is designed to provide a combination of all of these.

Unfortunately, the amplifier is not a perfect device. There are several major imperfections that system designers need to worry about when using amplifiers in a system.

- First, an amplifier introduces noise, in addition to providing gain.
- Second, the gain of the amplifier depends on the total input power. For high input powers, the EDFA tends to saturate and the gain drops. This can cause undesirable power transients in networks.
- Finally, although EDFAs are a particularly attractive choice for WDM systems, their gain is not flat over the entire passband. Thus some channels see more gain than others. This problem gets worse when a number of amplifiers are cascaded.

Gain Saturation in EDFAs

- An important consideration in designing amplified systems is the saturation of the EDFA.
- Depending on the pump power and the amplifier design itself, the output power of the amplifier is limited.
- As a result, when the input signal power is increased, the amplifier gain drops.



- For low input powers, the amplifier gain is at its unsaturated value, and at very high input powers, G →1 and the output power Pout = Pin.
- The output saturation power $Pout^{sat}$ is defined to be the output power at which the amplifier gain has dropped by 3 dB.

- The output saturation power is given as (assuming G=1 and $P_{out} = G.P_{in}$)

$$Pout^{sat} = P^{sat} \ln 2$$

The saturation power is a function of the pump power and other amplifier parameters. It is quite common to have output saturation powers on the order of 10 to 100 mW (10 to 20 dBm).

Crosstalk

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- Crosstalk is the general term given to the effect of other signals on the desired signal.
- Almost every component in a WDM system introduces crosstalk .

- The components include filters, wavelength multiplexers/demultiplexers, switches, semiconductor optical amplifiers, and the fiber itself (by way of nonlinearities).

- Two forms of crosstalk arise in WDM systems: interchannel crosstalk and intrachannel crosstalk

(i) Intrachannel Crosstalk

- Crosstalk signal is at the same wavelength as that of the desired signal or sufficiently close to it that the difference in wavelengths is within the receiver's electrical bandwidth. This form of crosstalk is called intrachannel crosstalk or, sometimes, coherent crosstalk. Intrachannel crosstalk effects can be much more severe than interchannel crosstalk

- Intrachannel crosstalk arises in transmission links due to reflections. This is usually not a major problem in such links since these reflections can be controlled. However, intrachannel crosstalk can be a major problem in networks. One source of this arises from cascading a wavelength demultiplexer (demux) with a wavelength multiplexer (mux),

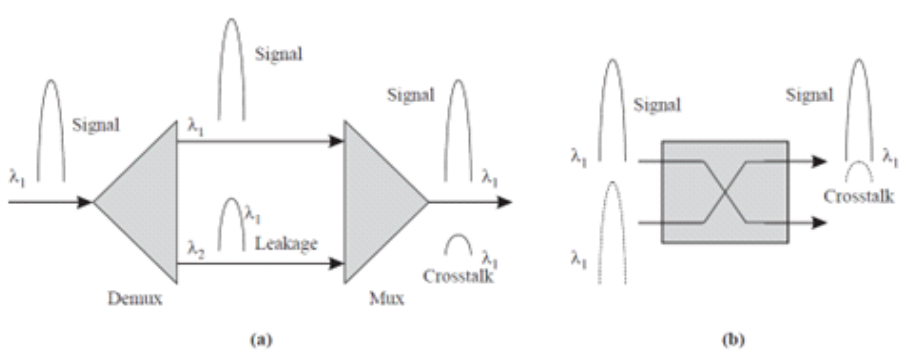

Figure: Sources of intrachannel crosstalk. (a) A cascaded wavelength demultiplexer and a multiplexer, and (b) an optical switch.

The demux ideally separates the incoming wavelengths to different output fibers. In reality, however, a portion of the signal at one wavelength, say, λi , leaks into the adjacent channel λi+1 because of nonideal suppression within the demux. When the wavelengths are combined again into a single fiber by the mux, a small portion of the λi that leaked into the λi+1 channel will also leak back into the common fiber at the output. Although both signals contain the same data, they are not in phase with each other, due to different delays they encounter. This causes intrachannel crosstalk. Another source of this type of crosstalk arises from optical switches, as shown in Figure (b), due to the nonideal isolation of one switch port from the other. In this case, the signals contain different data.

(ii) Interchannel Crosstalk

Crosstalk signal is at a wavelength sufficiently different from the desired signal's wavelength that the difference is larger than the receiver's electrical bandwidth. This form of crosstalk is called interchannel crosstalk. Interchannel crosstalk can also occur through more indirect interactions, for example, if one channel affects the gain seen by another channel, as with nonlinearities.

Interchannel crosstalk can arise from a variety of sources. A simple example is an optical filter or demultiplexer that selects one channel and imperfectly rejects the others, as shown in Figure (a). Another example is in an optical switch, switching different wavelengths (shown in Figure (b)), where the crosstalk arises because of imperfect isolation between the switch ports.

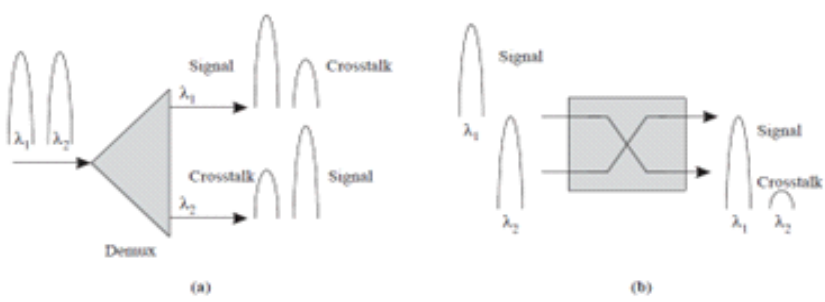

Figure: Sources of interchannel crosstalk. (a) An optical demultiplexer, and (b) an optical switch with inputs at different wavelengths.

Figure below shows the crosstalk penalties plotted against the crosstalk level for intrachannel and interchannel crosstalk, which we will consider next. If 1 dB penalty is allowed with signal-independent noise, then the intrachannel crosstalk level should be 20 dB below the desired signal

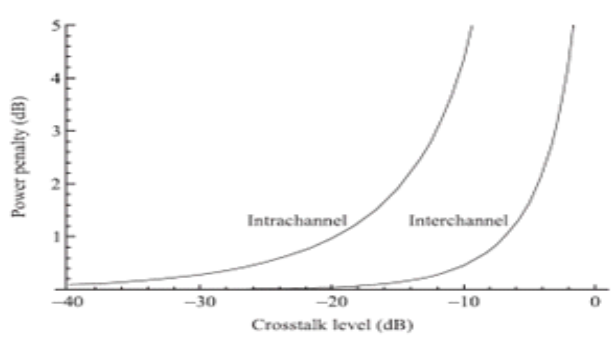

Figure: Intrachannel and interchannel crosstalk power penalties that are limited by thermal noise are shown as a function of crosstalk level

Network Management Functions

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Classically, network management consists of several functions, all of which are important to the operation of the network:

1. Performance Management:

- Performance management deals with monitoring and managing the various parameters that measure the performance of the network.
- Performance management is an essential function that enables a service provider to provide quality-of-service guarantees to their clients and to ensure that clients comply with the requirements imposed by the service provider.
- It is also needed to provide input to other network management functions, in particular, fault management, when anomalous conditions are detected in the network.

2. Fault management

- This is responsible for detecting failures when they happen and isolating the failed component.
- The network also needs to restore traffic that may be disrupted due to the failure, but this is usually considered a separate function.

3. Configuration Management

- Configuration management deals with the set of functions associated with managing orderly changes in a network.
- The basic function of managing the equipment in the network belongs to this category.
- This includes tracking the equipment in the network and managing the addition/removal of equipment, including any rerouting of traffic this may involve and the management of software versions on the equipment.
- Another aspect of configuration management is connection management, which deals with setting up, taking down, and keeping track of connections in a network.
- This function can be performed by a centralized management system. Alternatively, it can also be performed by a distributed network control entity.

4. Security Management

- Security management includes administrative functions such as authenticating users and setting attributes such as read and write permissions on a per-user basis.
- From a security perspective, the network is usually partitioned into domains, both horizontally and vertically.
- Vertical partitioning implies that some users may be allowed to access only certain network elements and not other network elements.
- Horizontal partitioning implies that some users may be allowed to access some parameters associated with all the network elements across the network

5. Accounting Management

- Accounting management is the function responsible for billing and for developing lifetime histories of the network components.
- This function is the same for optical networks, compared to other networks.

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- The semiconductor lasers used in optical communication systems are relatively low-power devices; nevertheless, their emissions can cause serious damage to the human eye, including permanent blindness and burns.
- The closer the laser wavelength is to the visible range, the more damage it can do, since the cornea is more
- transparent to these wavelengths.
- For this reason, systems with lasers must obey certain safety standards. Systems with lasers are classified according to their emission levels, and the relevant classes for communication systems

1. **Class I system:**

- A Class I system cannot emit damaging radiation. The laser itself may be a high-power laser, but it is prevented from causing damage by enclosing it in a suitably interlocking enclosure.
- The maximum power limit in a fiber for a Class I system is about 10 mW (10 dBm) at 1.55 µm and 1 mW(0 dBm) at 1.3 µm.
- Moreover, the power must not exceed this level even under a single failure condition within the equipment. A typical home CD player, for example, is a Class I system.

2. **Class IIIa System:**

- A Class IIIa system allows higher emission powers—up to 17 dB min the 1.55 µm wavelength range—but access must be restricted to trained service personnel.
- Class IIIa laser emissions are generally safe unless the laser beam is collected or focused onto the human eye.

3. **Class IIIb System:**

A Class IIIb system permits even higher emission powers, and the radiation can cause eye damage even if not focused or collected.

Under normal operation, optical communication systems are completely "enclosed" systems—laser radiation is confined to within the system and is not seen outside.
The problem arises during servicing or installation, or when there is a fiber cut, in which case the system is no longer completely enclosed and emission powers must be kept below the levels recommended for that particular system class.
Communication systems deployed in the enterprise world must generally conform to Class I standards since untrained users are likely to be using them.
Systems deployed within carrier networks, on the other hand, may likely be Class IIIa systems, since access to these systems is typically restricted to trained service personnel.

The safety issue thus limits the maximum power that can be launched into a fiber. For single-channel systems without optical power amplifiers using semiconductor lasers, the emission levels are small enough (−3 to 0 dBm typically) that we do not have to worry much about laser safety.

However, with WDM systems, or with systems using optical power amplifiers, we must be careful to regulate the total power into the fiber at all times.

Optical Layer Services and Interfacing

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The optical layer provides lightpaths to other layers such as the SONET/SDH, IP/MPLS, and Ethernet layers, as well as the electronic layer of the Optical Transport Network (OTN), which includes the the optical channel transport unit (OTU) and optical channel data unit (ODU) sublayers . In this context, the optical layer can be viewed as a server layer, and the higher layer that makes use of the services provided by the optical layer as the client layer. From this perspective, we need to specify clearly the service interface between the optical layer and its client layers. The key attributes of such a managed lightpath service are the following:

- Lightpaths need to be set up and taken down as required by the client layer and as required for network maintenance.
- Lightpath bandwidths need to be negotiated between the client layer and the optical layer. Typically, the client layer specifies the amount of bandwidth needed on the lightpath.
- An adaptation function may be required at the input and output of the optical network to convert client signals to signals that are compatible with the optical layer. This function is typically provided by transponders, The specific range of signal types, including bit rates and protocols supported, need to be established between the client and the optical layer.
- Lightpaths need to provide a guaranteed level of performance, typically specified by the bit error rate
- Multiple levels of protection may need to be supported, , protected, unprotected, and protected on a best-effort basis.
- in addition to being able to carry low-priority data on the protection bandwidth in the network. In addition, restoration time requirements may also vary by application.
- Lightpaths may be unidirectional or bidirectional. Almost all lightpaths today are bidirectional. However, if more bandwidth is desired in one direction compared to the other, it may be desirable to support unidirectional lightpaths.
- A multicasting, or a drop-and-continue, function may need to be supported. Multicasting is useful to support distribution of video or conferencing information. In a drop-and-continue situation, a signal passing through a node is dropped locally, but a copy of it is also transmitted downstream to the next node.
- Jitter requirements exist, particularly for SONET/SDH connections. In order to meet these requirements, 3R regeneration may be needed in the network. Using 2R regeneration in the network increases the jitter, which may not be acceptable for some signals.
- There may be requirements on the maximum delay for some types of traffic. This may place restrictions on maximum allowed propagation delay (or equivalent link length) on links. This will need to be accounted for while designing the lightpaths.

Extensive fault management needs to be supported so that root-cause alarms can be reported and adequate isolation of faults can be performed in the network. This is important because a single failure can trigger multiple alarms. The root cause alarm reports the actual failure, and we need to suppress the remaining alarms. Not only are they undesirable from a management perspective, but they may also result in multiple entities in the network reacting to a single failure, which cannot be allowed.

Enabling the delivery of these services requires a control and management interface between the optical layer and the client layer. This interface allows the client to specify the set of lightpaths that are to be set up or taken down and set the service parameters associated with those lightpaths. The interface also enables the optical layer to provide performance and fault management information to the client layer. This interface can take on one of two facets. The simple interface used today is through the management system. A separate management system communicates with the optical layer EMS, and the EMS in turn then manages the optical layer.

Introduction to free space optics

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Spectrum scarcity, coupled with bandwidth appetites in metropolitan area networks (MANs), is forcing wireless operators and service provides to look at new methods to connect cells. They are faced with many options in attempts to meet the high bandwidth demand.

- The first and often times most obvious choice is fiber-optic cable, but the associated delays and costs to lay fiber often make it economically prohibitive.
- The second alternative is radio frequency (RF) technology, which is a mature technology for longer distance transmission than FSO, but RF technologies cannot scale to optical capacities of 2.5 Gbps. The current RF bandwidth ceiling is 622 Mbps.
- The third option is copper-based technologies such as T1, cable modem, or DSL. Although copper infrastructure is available almost everywhere, it is still not a viable alternative because the bandwidth limitation of 2 to 3 Mbps makes it a marginal solution.
- The fourth and most viable choice is FSO. The technology is an optimal solution, given its optical base, bandwidth scalability, speed of deployment, portability, and cost-effectiveness

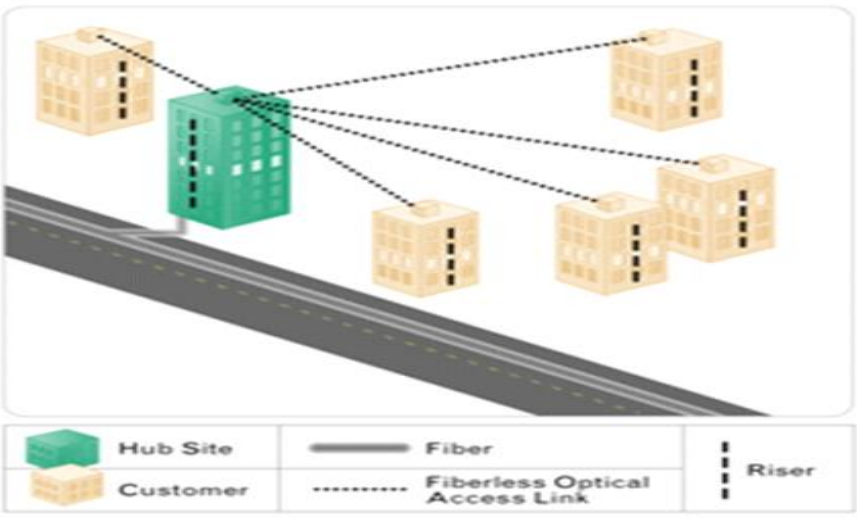


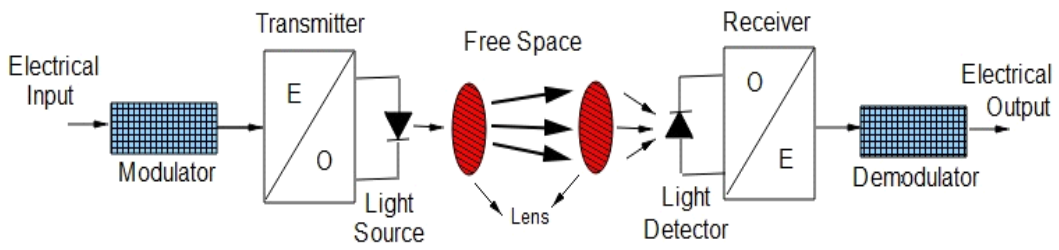
Table 1 Comparision of FSO with other access technologies (Buckley, 2001).

Features	FSO	Fiber	DSL
Deployment time	Days to weeks	4-12 months	6-12 months
Provisioning time	Immediate	Complex	Complex
Initial investment for few subscribers	Low	High	High
Reliability	Medium	High	High
Topology/flexibility	PP, PM, Mesh	PP, PM, Mesh	PP
Distance Limitation	200 – 2000 m	200 km	5.5 km
Bandwidth/speed	1.25 Gbps	10 Gbps	2 Mbps

Free-space optics (FSO), also known as fiber-free or fiberless photonics, refers to the transmission of modulated light pulses through free space (air or the atmosphere) to obtain broadband communications. Laser beams are generally used, although non-lasing sources such as light-emitting diodes (LEDs) or IR-emitting diodes (IREDs) will serve the same purpose. FSO can be the best wireless solution where fiber optical cable is not available, high bandwidth (anywhere from 1 Mbps up to 1.25 Gbps) is required, and line-of-sight can be obtained to a target within a couple of miles

Elements of FOS:

Transmission using FSO technology is relatively simple. It involves two systems each comprising of an optical transceiver which consists of a laser transmitter and a receiver to provide full duplex (bi-directional) capability.



Electrical input is a network traffic into pulses of invisible light representing 1`s and 0`s. The transmitter, which consists of two part main parts: an interface circuit and source driver circuit, converts the input signal to an optical signal suitable for transmission. The drive circuit of the transmitter transforms the electrical signal to an optical signal by varying the current follow through the light source. Transmitter function is to project the carefully aimed light pulses into the air. The information signal modulates the field generated by the optical source. The modulated optical field then propagates through a free-space path before arriving at the receiver.

In the receiver side, transmitted data realizes inverse operations i.e., photo detector converts the optical signal back into an electrical form as indicated in previous figure. In other words, a receiver at the other end of the link collects the light using lenses and/or mirrors. Received signal converted back into fiber or copper and connected to the network. Reverse direction data transported the same way (full duplex).

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Advantages

- It operates in a completely unregulated frequency spectrum (range of THz). Because there is little or no traffic currently in this range, the FCC has not required licenses above 600 GHz. This means FSO is not likely to interfere with other transmissions.
- Cost is a major advantage. Significantly lower cost on average than the construction of a new fiber optical solution, or leased lines. With FSO, there is also no capital overhang.
- It can be deployed in days to weeks versus months to years (no excavation of sidewalks, building permits, etc.)
- Bandwidth can easily be scaled with virtually unlimited headroom (10 Mbps to 1.25 Gbps) per link.
- An FSO network architecture need not be changed when other nodes (buildings) are added; customer capacity can be easily increased by changing the node numbers and configurations.
- The technology is easily upgradeable, and its open interfaces support equipment from a variety of vendors, which helps carriers protect the investment in their embedded infrastructures

Challenges

Despite its potential, FSO still has many challenges to overcome before it will be deployed widely. These challenges includes,

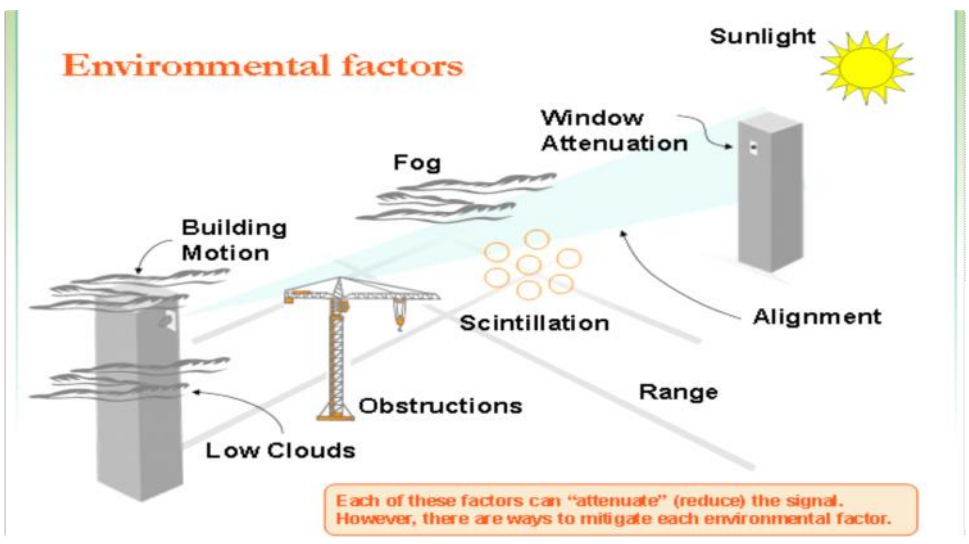
1.Weather Conditions: FSO is an LOS technology, meaning that nodes must have an unobstructed path to the hub antenna. In other words, interference of any kind can pose problems. One form of interference is the weather condition. A major challenge to FSO communications is fog.

As shown in Table, there could be to 300 dB per kilometer of attenuation on foggy days. Snow or rain does not bother FSO much, because the light can find its way through the raindrops and snowflakes, but fog can stop lasers like a brick wall. Fog is vapor composed of water droplets, which are only a few hundred microns in diameter but can modify light characteristics or completely hinder the passage of light through a combination of absorption, scattering and reflection.

Weather Conditions	Attenuation
Clear	5 to 15 dB/km
Rain	20 to 50 dB/km
Snow	50 to 150 dB/km
Fog	50 to 300 dB/km

2. Absorption:

Absorption occurs when suspended water molecules in the terrestrial atmosphere extinguish photons. This causes a decrease in the power density (attenuation) of the FSO beam and directly affects the availability of a system. Absorption occurs more readily at some wavelengths than others.



3. Scattering:

Scattering is caused when the wavelength collides with the scatterer like flying birds, building, structures, and towers. The physical size of the scatterer determines the type of scattering. Unlike absorption, there is no loss of energy in scattering, only a directional redistribution of energy that may have significant reduction in beam intensity for longer distances

4. Scintillation:

Heated air rising from the earth or man-made devices such as heating ducts creates temperature variations with different air pockets. This can cause fluctuations in signal amplitude which leads to "image dancing" at the FSO receiver end. One way of minimizing the effect of scintillation is to have no links longer than 500m and mount systems farther away from vents, hot roofs, and air-conditioning ducts.

5. Building Motion

The building motion due to structural vibrations, tilts gives attenuation of light signal. Table below shows different factors, cause and magnitude of Attenuation.

Type	Cause(s)	Magnitude	Frequency
Tip/tilt	Thermal expansion	High	Once per day
Sway	Wind	Medium	Once every several seconds
Vibration	Equipment (e.g., HVAC), door slamming, etc.	Low	Many times per second

6. Safety:

The two major concerns typically expressed involve questions about human exposure to laser beams (which present much more danger to the eyes than any other part of the human body) and high voltages within the laser systems and their power supplies. Several standards have been developed covering the performance of laser equipment and the safe use of lasers.

Applications

- Metro Area Networks (MAN)
- Last Mile Access
- Enterprise connectivity
- Fiber backup
- Backhaul
- Service acceleration